

Battery Health (SOH & RUL) Prognostics

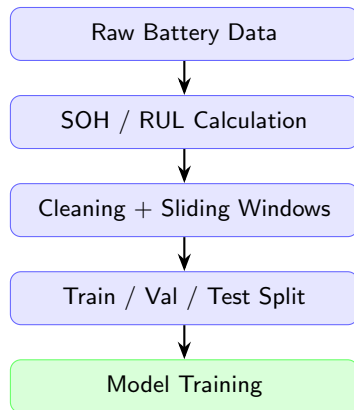
From Machine Learning to Deep Learning

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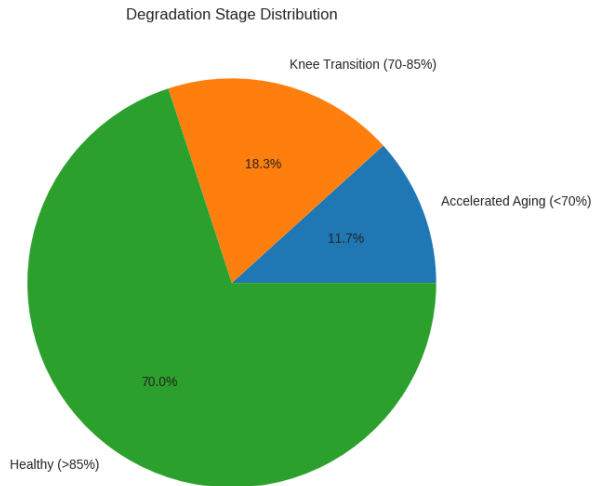
Project Overview & Data Pipeline

- Took raw battery cycle data
- Computed **SOH** (State of Health) and **RUL** (Remaining Useful Life) for each cycle
- Cleaned the data and created sequences (windows)
- Performed Train / Validation / Test split **by battery** (to avoid data leakage)
- Trained all models on this same clean dataset



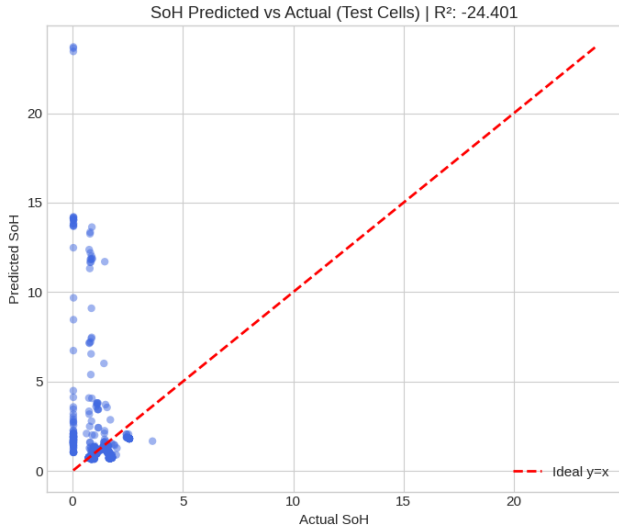
Battery Degradation Analysis

- Battery degrades over time
- 3 stages were observed:
 - **Healthy** ($>85\%$ SOH)
 - **Knee Transition** (70–85%)
 - **Accelerated Aging** ($<70\%$)
- Most cycles fall in the **Healthy** stage
- Only $\sim 12\%$ of cycles are in the critical (Accelerated Aging) zone



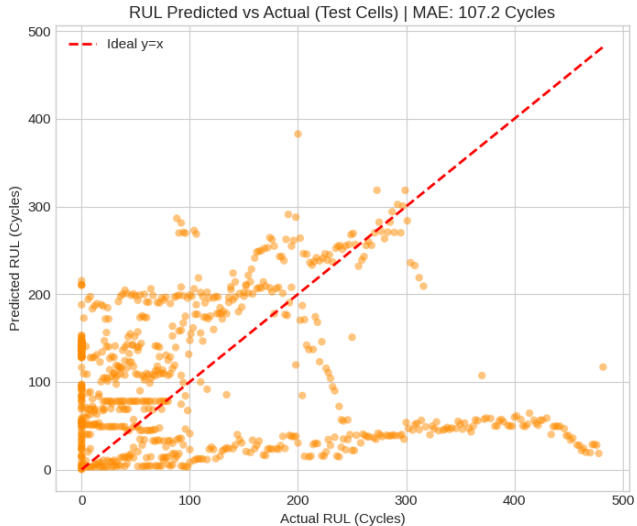
Baseline Models: SoH Prediction (Random Forest)

- First, simple baseline models were tried
- **Ridge Regression** – a simple linear model
- **Random Forest** – a somewhat stronger, non-linear model
- This graph shows SoH predictions on unseen test batteries
- Points are quite far from the red (ideal) line – meaning predictions are weak



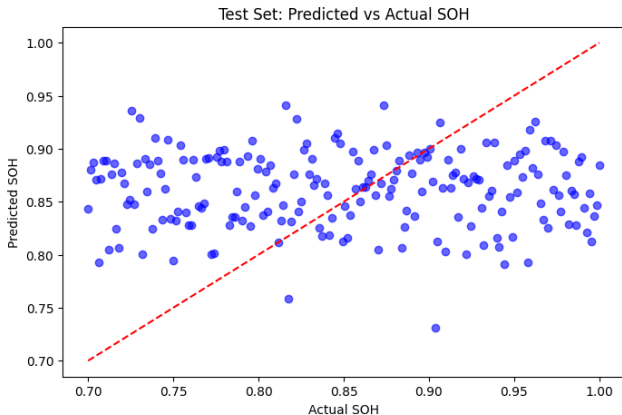
Baseline Models: RUL Prediction (Random Forest)

- The same baseline model was also used to predict **RUL** (Remaining Useful Life)
- Mean Absolute Error ≈ 107 cycles – quite high
- Predictions are very scattered and don't follow the ideal line
- This is why Deep Learning models were needed



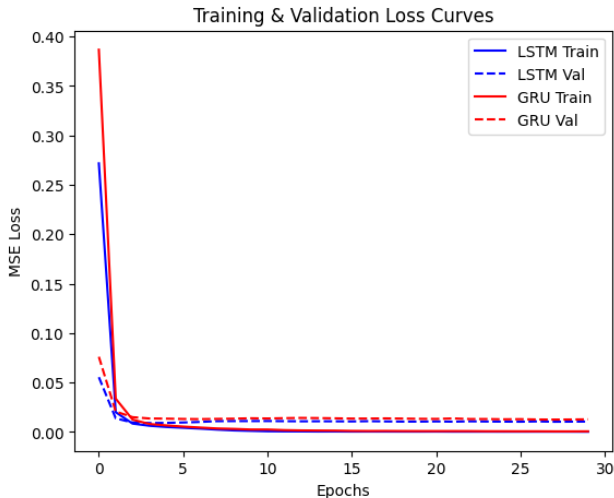
Deep Learning: LSTM for SOH Prediction

- **LSTM** (Long Short-Term Memory) works best for sequence data
- Predicts the next SOH by looking at the battery's past cycles
- Results were considerably better than the baseline models
- Predicted values came very close to the actual SOH
- This was our first deep learning model



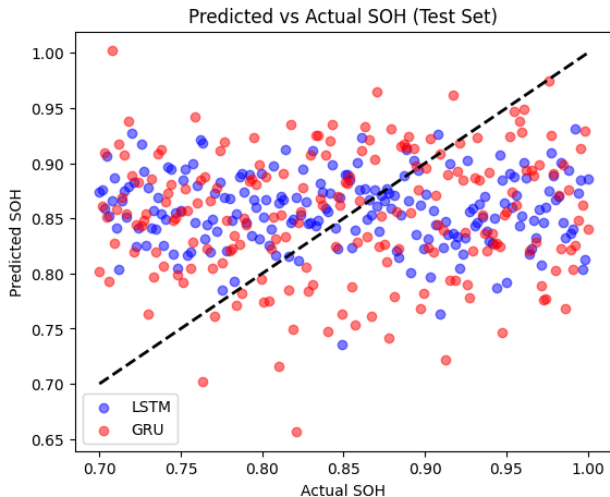
LSTM vs GRU: Training & Validation Loss

- **GRU**: similar to LSTM, but simpler and faster
- Both models' loss dropped very quickly (within a few epochs)
- Train and Validation loss curves are close – overfitting is low
- This is a sign of good convergence



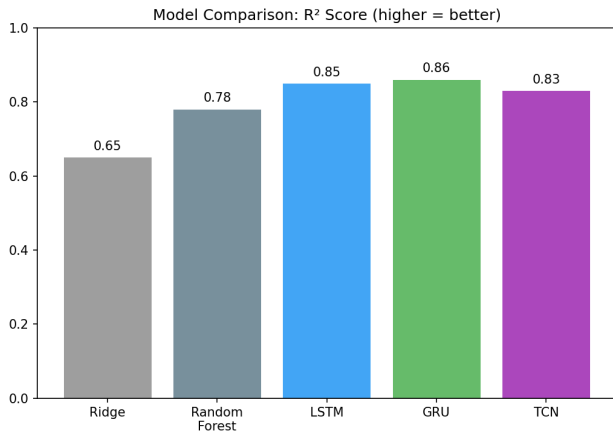
LSTM vs GRU: Predicted vs Actual SOH

- Both models' predictions are shown together in this graph
- **GRU** (red) gave overall slightly better accuracy
- **LSTM** (blue) also gives fairly close results
- Both models are considerably better than the baseline



Final Comparison: Accuracy (R^2 Score)

- **Ridge/RF**: Least accurate (R^2 0.65–0.78)
- **LSTM, GRU, TCN**: All $R^2 > 0.8$
- **GRU** is the best – $R^2 = 0.86$
- Deep learning models improved significantly over the baseline



Final Comparison: Error (MAE) & Conclusion

- **GRU** has the lowest error (MAE = 0.0279)
- **TCN** trains fastest, with low error as well
- **Conclusion:** GRU = best accuracy, TCN = best speed
- **Next Step:** Hyperparameter tuning on GRU and TCN

